FACULTY
OF MATHEMATICS
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To whom it may concern

Prague, September 30, 2026

Doctoral Thesis of Jiří Klepl – Supervisor's Reference

This doctoral thesis focuses on performance optimization of scientific algorithms by improving their memory access patterns. A key aspect is the design of layout-agnostic and traversal-agnostic abstractions that decouple memory-related optimizations from algorithm design. With this agnostic approach, programmers, who are usually domain experts, can focus on expressing their intended algorithms, while memory optimizations can be left to software engineering researchers, autotuning tools, or even AI models.

The thesis is organized as a collection of published papers, accompanied by an introduction that explains the agnostic principle and places the research outcomes in a broader context. The main part covers the design of agnostic abstractions. The author devised a way to express both the layouts and traversal orders of regular data structures, such as grids or tensors, as simple building blocks implemented as C++ classes. Thanks to C++ template metaprogramming, the assembly of final structures and traversal loops is performed at compile time, resulting in a zero-overhead abstraction. In addition, the author addresses important related concepts, such as traversal parallelization using both CPU libraries and CUDA for GPU offloading, MPI integration for distributed computing with native translation between internal data structures and MPI types, and an interface that simplifies autotuning with existing tools. Beyond the publications themselves, this line of research also yielded Noarr, an open-source header-only library that is publicly available on GitHub.

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The final part of the research shifts attention to LLM-assisted code design and optimization. LLMs have advanced tremendously since Mr. Klepl entered his Ph.D. program, so this part of the research was not outlined in advance; however, it appears to be a logical extension. This research helped identify the limitations of LLMs in the domain of HPC optimization and explored possibilities for integrating LLMs with existing abstractions, both those developed in this thesis and those provided by other libraries.

Furthermore, the author engaged in collaborative research with Matyáš Brabec, a Ph.D. student from the same research group. Mr. Brabec focuses on scientific simulations implemented as stencil algorithms. Together, they were able to apply the knowledge gained from agnostic abstractions and C++ template metaprogramming to design Cellato, a framework that simplifies the implementation of cellular automata (CA) simulations. In Cellato, the author needs only to encode CA rules in a DSL based on C++ templates; the simulation itself is then automatically parallelized for either CPU or GPU execution, using efficient bit-plane encoding and bitwise vectorization.

During his Ph.D. studies, Jiří Klepl demonstrated that he is capable of conducting scientific research both independently and as part of a team. He published regularly and took part in other academic duties in accordance with his study plan.

In summary, I strongly recommend the thesis for defense.

Martin Kruliš
(supervisor)